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The cognitive neuroscience of self-awareness: Current framework, clinical implications, and future research directions

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Abstract

Self-awareness, the ability to take oneself as the object of awareness, has been an enigma for our species, with different answers to this question being provided by religion, philosophy, and, more recently, science. The current review aims to discuss the neurocognitive mechanisms underlying self-awareness. The multidimensional nature of self-awareness will be explored, suggesting how it can be thought of as an emergent property observed in different cognitive complexity levels, within a predictive coding approach. A presentation of alterations of self-awareness in neuropsychiatric conditions will ground a discussion on alternative frameworks to understand this phenomenon, in health and psychopathology, with future research directions being indicated to fill current gaps in the literature.

This article is categorized under:

Philosophy > Consciousness

Psychology > Brain Function and Dysfunction

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KEYWORDS

agency, interoception, metacognition, neuropsychiatric conditions, self-awareness

1 | INTRODUCTION

Self-awareness can be defined as the ability to take oneself as the object of awareness (Morin, 2011). Although this definition may suggest the idea of a unitary self that acts as both observer and object of awareness, in reality there is a multiplicity of self-processes included in this self-model, with different features being at the focus of awareness at any given time (Mograbi et al., 2021). Accordingly, self-awareness can be thought of not as a single unified capacity, but rather as a variety of processes (Mograbi et al., 2021). These would include, for example, both the representation of internal bodily states (i.e., interoception) and declarative personal information about oneself (i.e., autobiographical

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memory). It follows from this multiplicity of self-processes that self-awareness can be differentially affected by clinical conditions, with some features, but not others, showing alterations.

The study of self-awareness has a number of important implications. Clinically, a better understanding of alterations in self-awareness in neurological and psychiatric conditions may foster advances in psychosocial (e.g., neuropsychological rehabilitation in dementia, Marinho et al., 2020; metacognitive training in psychosis, Lysaker et al., 2018) and pharmacological interventions (e.g., psilocybin for depression; Goodwin et al., 2022). Theoretically, considering that cognition emerged, in an evolutionary perspective, for self-serving purposes, elucidating the mechanisms of self-awareness may lead to a finer comprehension of brain functioning in general. Finally, a cognitive neuroscience of self-awareness may inform our understanding of the extent to which different patient groups, extant species and devices are capable of self-awareness, with important legal and ethical implications.

Despite this relevance, there have been few attempts to offer an integrative framework to different self-awareness processes (e.g., Morin, 2017), identifying underlying mechanisms and suggesting potential interactions between them. Critically, an overview of both neurocognitive mechanisms and an integration of those with alterations in different clinical conditions is still lacking in the literature. Such an approach can lead to theoretical and clinical advances.

Accordingly, the current review aims to discuss the neurocognitive mechanisms underlying self-awareness. For this purpose, a discussion of the multidimensional nature of self-awareness will be followed by a review of how self-awareness can be considered as an emergent property observed across different complexity levels. A discussion of alterations of self-awareness in neuropsychiatric conditions will highlight some of the mechanisms discussed previously, as well as indicating the involvement of self-awareness in psychopathology and health. Alternative frameworks for understanding self-awareness will be discussed, followed by a blueprint for future research.

2 | COGNITIVE NEUROSCIENCE OF SELF-AWARENESS: MULTIPLE DIMENSIONS AND COMPLEXITY LEVELS

2.1 | Self-awareness processes

As indicated above, self-awareness should not be considered a unified capacity, but rather a collection of processes related to different self-functions. These processes are linked to different objects of awareness (Marková, 2006), that is, specific foci that dominate subjective awareness at any given time. Here, a basic, if schematic, distinction between two sources of information may be relevant: (1) self-awareness abilities linked to bodily processes, such as visceral states (interoception), body position/ownership (proprioception) and action (agency); (2) capacities related to a representational level, describing our cognitive and emotional abilities (respectively, metacognition and emotional regulation), as well as records of self-related information (autobiographical memory). This distinction does not imply an absence of information exchange between these sources; for example, prior experiences grounded by autobiographical memory may impact the perception of current interoceptive processes. As discussed below, for each self-awareness process there may be varying levels of cognitive complexity.

At its most basic level, self-awareness can be exemplified by awareness of internal visceral states, that is, *interoception*. Interoception can be defined as the sense of the physiological state of the body (Critchley, 2004). Potentially it represents one of the most basic aspects of self-representation (Quadt et al., 2018), being directly linked to the maintenance of living conditions for the organism. Although the bulk of this process may be done implicitly through homeostatic reflexes, awareness of bodily states and associated affective reactions may assist in allostatic responses. Information exchange between the brain and the body generate a self-representation, including physical needs, that would guide and motivate behavioral responses. Different dimensions of interoception have been identified, such as accuracy (performance on objective visceral detection), sensibility (self-evaluated assessment of subjective interoception) and awareness (confidency-accuracy correspondence; Garfinkel et al., 2015).

In addition to monitoring internal states, self-awareness is also linked to the creation of a bodily ensemble. Experiencing our body as belonging to ourselves is referred to as *body ownership*. It emerges from multisensory integration (Chancel & Ehrsson, 2020), being closely linked, but not reduced, to *proprioception*, the sense of relative position of different body parts. It is a feature essential for survival, by establishing boundaries between internal (self) and external environments. The notion that body ownership is not an a priori, but rather the result of a process, becomes particularly clear in cases of its disruption, for example in specific clinical conditions (e.g., alien hand syndrome; Hassan & Josephs, 2016) and with the action of certain substances (e.g., ketamine; Margulho et al., 2022).

The body acts in the environment, and discerning which actions have been caused by ourselves, *agency*, is a fundamentally adaptive feature. Agency is closely linked to action generation and the feeling of voluntary control (Haggard & Chambon, 2012). Models of agency typically rely on the concept of efference copy, which has its roots in corollary discharge theory (Feinberg, 1978). According to this perspective, an intention to move leads not only to a motor command, but also to a copy of this command (the efference copy) to predict the consequences of action. When action prediction and feedback of action match, feeling of agency is experienced.

Our brains create self-models not only for our actions, but also for our cognitive and emotional capabilities. Based on previous information and expectations, metacognition involves the development of cognitive strategies, including the allocation of sufficient cognitive resources in order to succeed in the execution of tasks (Rhodes, 2019). It is a predictor of awareness of performance and the ability to appraise our capacities, representing important features of our model of self, including self-efficacy and self-regulated learning (Rhodes, 2019). It can be considered a higher-order self-awareness process, given that it deals primarily with complex representations and has, by definition, a recursive nature.

In a similar way, *emotional regulation* describes thoughts, strategies, and responses in relation to emotional processes, such as arousal, feelings, and affect-laden cognition. Similar to other self-awareness processes, there are implicit mechanisms for emotional regulation, such as parasympathetic responses after intense arousal. Nevertheless, as with metacognition, emotional regulation implies a relatively higher-order processing and manipulation of information linked to goal-directed behavior. An influential model of emotional regulation divides its processes into antecedent-focused and response-related (Gross, 2002). The former includes selection of situations (e.g., avoidance, disengagement), modification of situations (e.g., problem solving, search for social support), attentional allocation (e.g., distraction, mindfulness, focusing on positive features) and cognitive reappraisal (e.g., humor, acceptance, threat appraisal), while the latter involves response suppression, aggression, and substance abuse, among others (Gross, 2002).

These evaluative abilities rely, crucially, on records of our previous experiences and trait knowledge about ourselves. *Autobiographical memory* is the ability to recall memories that refer to one's personal history. It has been traditionally divided into two types of memory: incident memory, the recollection of specific episodes from an individual's life ("I remember one day at university ..."); and personal semantics, general information about oneself ("I studied at X university"). It is possible to consider episodic and semantic memory as part of a continuum (Burianova et al., 2010), with some information clearly being closer to one of the poles, but most lying in between these two types of record. This distinction remains useful, nevertheless, given that different neural correlates have been found for memories depending on their level of episodic retrieval (Burianova & Grady, 2007; Cabeza et al., 2002), which may have clinical implications, as discussed below.

It is important to note that the self-awareness abilities covered here may not represent an exhaustive list. Focus was given to processes with a clear neurobiological basis, also considering an evolutionary perspective, that is, the presence of some of these abilities, albeit in less complex forms, in other species. Other fields of research have approached self-awareness from different perspectives. For instance, the social/personality psychology approach has privileged concepts such as self-esteem and self-reflection (for a glossary of terms, see Morin, 2017). Although these are certainly relevant, it is possible that they represent outcomes of more basic self-awareness processes, and not their bases.

Similarly, inner speech has been highlighted as an important feature of self-awareness (Morin, 2018). Inner speech would allow identifying, organizing, consolidating, questioning, and retrieving information pertaining to oneself (Morin, 2018). While it is undeniable that we, as humans, live in a world of narrative, and that internal dialogue may modulate self-awareness, it is also possible to consider the existence of models of self and self-awareness without narrative, for example, in other species, infants and severe cases of neurodegenerative conditions. Inner speech may be important in higher-order self-awareness processes, but, as above, it can also be considered as the result of such processes, for instance with the outcomes of metacognition and emotional regulation feeding into the internal dialogue ("I'm capable," "I'm nervous right now").

The focus chosen here on self-awareness also prevents a full exploration of other, related, concepts, such as self-reference, self-reflection or rumination (Trapnell & Campbell, 1999). Although these may be particularly relevant from a clinical perspective, with excessive self-reference and rumination being the hallmarks of depression (Hamilton et al., 2015) and self-reflection potentially being linked to benign interventions, such as mindfulness (Morin, 2018), again they may represent higher-order results of more basic self-awareness processes.

2.2 | Neurocognitive mechanisms of self-awareness

A crucial notion in relation to self-awareness is that it can be explored at different levels of cognitive complexity, from neuronal electrical activity to self-reported responses. According to this view, self-awareness is not a binary, all or none,

ability, being best conceptualized as a graded phenomenon. This suggests that self-awareness has deep biological roots, being ingrained in our basic cognitive activity, but also that, during the course of evolution, some of these mechanisms grew increasingly more elaborate. This may be particularly relevant to understand evidence for self-awareness across different species. Even the simplest organisms are capable of self-monitoring for regulatory purposes, but more complex nervous systems potentially allow higher-order, recursive, self-awareness abilities (Lage et al., 2022).

This is in line with the notion, indicated above, that most self-awareness abilities have their root in implicit self-monitoring processes. While this is more obvious in the case of bodily self-awareness, for example with implicit homeostatic reflexes coordinating basic interoceptive control (Quadt et al., 2018) even higher-order processes can also be considered under a similar framework. For instance, metacognition may rely, at its core, on basic monitoring at the electrophysiological level, for example with event-related potentials linked to the detection and reactivity to errors (Dehaene et al., 1994; Gehring et al., 1993). The support of self-awareness on lower-level implicit processes may be an important feature to understand this phenomenon from evolutionary and cognitive perspectives.

The point at which these self-monitoring abilities lead to subjective awareness remains discussed. One perspective to understand self-awareness would be to consider it as an example of an emergent property. According to this view, interactions at a lower level lead to new properties that are greater than the sum of their parts, and to which a specific identity must be recognized (Artibe & De Domenico, 2022; McClelland et al., 2010). Within this perspective, implicit sources of information would interact and be integrated into explicit, aware, self-models. The notion of emergence not only prevents reifying self-awareness, indicating it is a process, rather than a thing in our brains, but also avoids the idea of an infinite regress in relation to selfhood. If more complex levels are but the result of the interaction of lower, less complex levels, the capacities at each level are neither replicated above or below. With decreasing levels of cognitive complexity, subjective experience is lost at some point of the reduction (Dennett, 2014).

Emergence has been suggested early on as a mechanism for consciousness (e.g., Sperry, 1969). It is intrinsically linked to the established concept of varying levels of cognitive complexity within and across nervous systems, which is supported both by neuroanatomy and evolutionary theory (Lage et al., 2022), and also the idea that bottom-up information is organized in increasingly complex ensembles. In this sense, it may fit well with an important framework to understand cognition, and particularly self-awareness, namely predictive coding (PC).

PC departs from the notion of the brain as an inference machine, which has its roots in the field of perception studies, including the work of Helmholtz (von Helmholtz, 1962; Westheimer, 2008) and Gregory (Gregory, 1972). Faced with pressures for survival in an uncertain environment, the brain creates models trying to predict the causes of afferent information. One of the main cognitive tasks within this perspective is to minimize the mismatch between top-down descending predictions and the incoming bottom-up sensory information. Mismatches, or prediction errors, can be minimized either by adjusting prediction, for instance changing beliefs and correcting prior information, or the incoming external information, for example through action on the world (Hohwy, 2017). There has been growing evidence for the general ideas of this framework, both from empirical studies (e.g., Dijkstra et al., 2020; Kok et al., 2012) and computational models (e.g., Lotter et al., 2020; Rao & Ballard, 1999), with the suggestion that PC emerges from the need of energy efficiency within the physical constraints of the brain (Ali et al., 2021).

This framework has been used to explain self-awareness processes, and may be particularly relevant to better understand their interactions. In the context of interoception, it has been suggested that afferent information from visceral sources are compared to higher-order predictions about physiological states (Quadt et al., 2018). Prediction errors would lead to the adjustment of priors and beliefs about interoception, as well as efferent autonomic regulation. Considering agency, comparisons between predicted movements and their consequences with an efference copy of the movement and incoming feedback from the body would be linked to the feeling of cognitive control of actions (Kannape & Blanke, 2012). Similar models can be inferred for other self-awareness abilities, with top-down predictions and ascending sensory information being compared, and adjustments to self-models or engagement with the world being made when mismatches are detected. In the case of declarative, higher-order processes, a similar dynamic may occur between meta (second-order) representations about the self and first-order self-related information.

Understanding self-awareness through this general framework may also be particularly helpful to explore interactions between its processes. For example, it has been suggested that interoceptive predictions may impact on explicit forms of agentic action control, with higher interoceptive awareness being potentially linked to increased action regulation for conscious processes (Marshall et al., 2018). Interoceptive sensations may also be at the core of emotional experiences, with the suggestion that categorization of physiological information leads to subjective experiences of emotions as discrete states (Feldman-Barrett et al., 2007). According to this view, top-down predictions would organize incoming

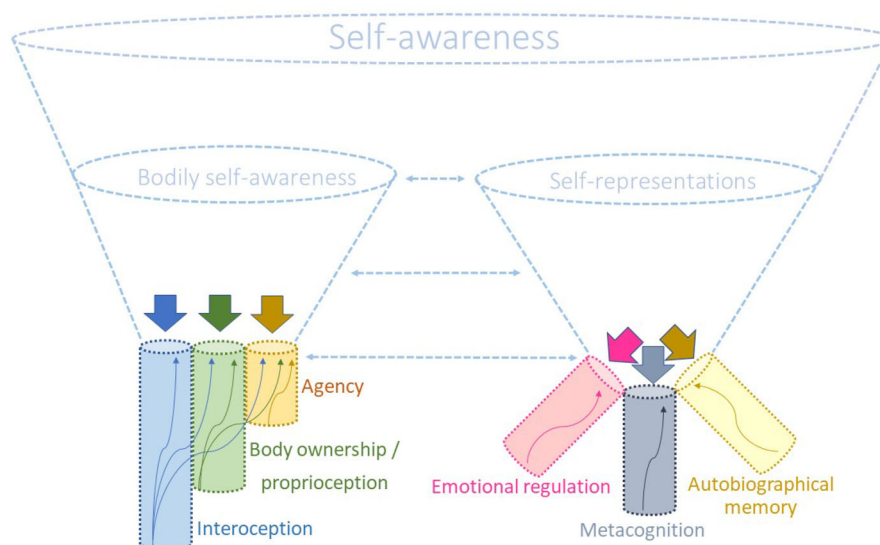


FIGURE 1 Integrated Self-Awareness Model (ISAM) - A schematic view of the interaction between self-awareness processes. Incoming bottom-up information (represented by thin arrows) is integrated into more complex cognitive levels and meet with top-down expectations and priors (thick arrows), leading to expression of different processes of self-awareness. Mismatches between those two sources of information result in adjustments in either of them or altered self-awareness. Bodily self-awareness emerges from the interaction of interoception, body ownership and agency, potentially with a nested hierarchy, in which interoceptive sensory signals also feed into the feeling of owning a body and agency. Higher-order processes such as emotional regulation, metacognition and autobiographical memory are also influenced by implicit bottom-up signals, but are expressed as explicit narrative self-representations, not necessarily with a hierarchical relationship between them. Bodily self-awareness and self-representations interact at the level of specific processes and higher integrative levels, leading to the emergence of an integrated sense of self-awareness.

visceral information into emotional concepts. Another example would be bottom-up electrophysiological error monitoring, in interaction with beliefs about performance, impacting on sense of agency and action competence. There are numerous potential relationships between processes and levels of self-awareness, leading to an integrative feeling of selfhood and self-awareness. Figure 1 presents a schematic view of some of these potential relations.

To summarize, the interaction between incoming sensory data from multiple modalities and top-down expectations/priors leads to the emergence of self-awareness. This property is, thus, the result of integration of information across sensory modalities, constrained by expectations and beliefs that exist as a consequence of recurrent experiences of the organism in its environment. The degree to which certain sources of information may be more relevant (e.g., interoceptive data) remains discussed, but it is possible that bodily experiences are at the core of any self-model, in a nested hierarchy approach (Feinberg, 2011). Understanding self-awareness from this perspective highlights both its multiplicity and dynamic nature. The focus here is on integration of information across processes and cognitive complexity levels, within a neurobiological approach, as opposed to previous attempts that have emphasized speech and narrative features of self-awareness (Morin, 2004; Morin, 2017).

3 | ALTERATIONS OF SELF-AWARENESS IN CLINICAL CONDITIONS

Here a brief overview of alterations in self-awareness across clinical conditions is presented. Far from being an exhaustive list, the conditions highlighted here are illustrative of the type of disruptions that can occur to self-awareness processes and also of the centrality of these alterations in the characterization of these disorders. Table 1 summarizes information about different self-awareness processes and potential conditions in which they are affected.

3.1 | Interoception

It is increasingly understood that abnormalities in interoception occur in several neuropsychiatric disorders, particularly conditions characterized by emotional disturbance (Nord & Garfinkel, 2022). In mood and anxiety disorders there

TABLE 1 Definitions about different self-awareness processes and clinical conditions in which they are potentially altered.

Self-awareness processes	Definition	Clinical conditions/symptoms
Interoception	Sense of the physiological conditions of the entire body	Alexithymia, anxiety disorders, autism spectrum disorders, depression, functional neurological disorders, schizophrenia, eating disorders
Proprioception and body ownership	Mapping of the relative position of body parts, and the feeling of owning a body, respectively	Functional movement disorders, schizophrenia, somatoparaphrenia, PTSD, depersonalization disorders
Agency	Sense of generating our own actions	Schizophrenia, Alien hand syndrome, anosognosia for hemiplegia
Metacognition	Monitoring, knowledge, and regulation of cognition	Dementia, obsessive-compulsive disorder, PTSD, psychosis
Emotional regulation	Monitoring and regulation of emotion	Borderline personality disorder, substance abuse disorder, bipolar affective disorder, autism spectrum disorders, ADHD
Autobiographical memory	Records of self-information, including specific episodes and general knowledge about oneself	Dementia, dissociative amnesia, posttraumatic stress disorder, depression, schizophrenia

is evidence of increased sensitivity and attention to interoceptive stimuli, that may also occur with reduced interoceptive accuracy (Bonaz et al., 2021). The increased attention to perceived threat seen in anxiety disorders may be associated with alterations in the interpretation or attribution of interoceptive signals, that further establishes anxious beliefs (Bonaz et al., 2021). This may be seen in the catastrophic interpretation of cardiovascular and other interoceptive signals as evidence of a heart attack in panic disorder, and the efficacy of cognitive and pharmacological treatments that aim to correct this misinterpretation (Nord & Garfinkel, 2022). Similarly, in depression, a negative view of the self may lead to, and be reinforced by, alterations in beliefs that determine the interpretation of interoceptive signals over the long term (Bonaz et al., 2021; Paulus & Stein, 2010). Impairment in interoceptive accuracy and awareness is also seen in functional neurological disorders (FND). In one study, people with functional seizures demonstrated lower interoceptive accuracy and greater subjective sensitivity to internal sensations than controls. The discrepancy between these measures, referred to as the interoceptive trait prediction error, was a strong predictor of dissociative symptoms (Koreki et al., 2020; Nord & Garfinkel, 2022).

People with schizophrenia also demonstrate evidence of reduced interoceptive accuracy (Monti et al., 2021). For example, in a study of 23 people with schizophrenia, interoceptive accuracy measured using a heartbeat perception task was significantly lower than controls (Ardizzi et al., 2016), and a recent transdiagnostic study also found evidence for pervasive interoceptive deficits in schizophrenia (Critchley et al., 2019). Impairments in interoception are associated with abnormalities in a network of brain regions including the insular cortex (Bonaz et al., 2021). A recent meta-analysis of neuroimaging studies including a range of interoceptive tasks and neuropsychiatric conditions found evidence of a common alteration in dorsal mid-insula activation, which may suggest a common role of interoceptive dysfunction across several mental health disorders (Nord et al., 2021).

3.2 | Proprioception and body ownership

Awareness of body ownership and the relative position of one's own body parts in space (proprioception) have been assessed using a number of experimental approaches, including the "rubber hand illusion" (RHI). In this paradigm, simultaneous touching of a participant's hand with a mannequin's hand may produce the illusion of body ownership of the fake hand (Rossetti et al., 2022). Studies using the RHI in people with schizophrenia have demonstrated abnormalities in the sense of body ownership, which correlate with the severity of hallucinations and delusions, passivity symptoms and anhedonia (Monti et al., 2021; Peled et al., 2003; Thakkar et al., 2011). The impairment of body ownership in schizophrenia may relate to deficits in multisensory integration or body-templates (Rossetti et al., 2022).

Traumatic experiences have also been associated with altered awareness of body ownership (Rabellino et al., 2018) and can occur in post-traumatic stress disorder (PTSD) and depersonalization disorders. In depersonalization, part of the body (e.g., a hand) can be experienced as "non-self," or there may be complete loss of body ownership where the

whole body is felt as non-self (Sierra & David, 2011). Depersonalization phenomena can occur in the context of neuropsychiatric conditions such as major depression or PTSD, or as a primary disorder (Medford et al., 2005). Depersonalization disorder (DPD) is characterized by a reduced sense of self, alteration in emotional experience and change in subjective first-person experience. A recent fMRI study in people with DPD demonstrated alterations in activation of the insula, also associated with interoception, highlighting the importance of this region in processes relating to bodily awareness (Medford et al., 2016).

3.3 | Agency

Agency, a sense of generating one's own actions, is notably affected in schizophrenia where sufferers often report the experience of their thoughts and/or actions being controlled by an external agent, symptoms known as thought interference and passivity phenomenon respectively (Walker et al., 2004). In healthy individuals, the sense of being the author of one's own actions has been found to be influenced via both conscious (explicit) and unconscious (implicit) routes (Aarts et al., 2005). To test the functioning of these two routes of influence, explicit and implicit, on agency in individuals with schizophrenia, Renes et al. (2013) designed an experiment where 23 people with schizophrenia were matched against 23 healthy controls in controlling the movements of a square on a computer screen. They found that, although both groups experienced agency when consciously determining the final resting place of the square, the schizophrenia group were significantly less likely to experience agency when the final resting place was unconsciously primed by 17-ms subliminal displays (Renes et al., 2013). These findings have been replicated by other researchers (Prikket et al., 2018), and this impaired sense of agency has also been shown to be present from very early on in the illness (Kozáková et al., 2020).

Conversely, anosognosic hemiplegic patients with damage to the right hemisphere experience a paradoxically increased sense of agency for movements performed by both plegic and healthy limbs (Cocchini et al., 2022). Increased perception of agency may also be behind pathological guilt, with excessive attribution of consequences to one's actions (Petrolini, 2020).

3.4 | Metacognition

Metacognition, or thinking about thinking, is an area of self-awareness that has been shown to be particularly affected in psychotic illnesses (Lysaker et al., 2013). A useful framework and tool for assessment of metacognition is the Metacognition Assessment Scale (MAS), which divides metacognition into three different domains: understanding one's own mind, understanding others' minds, and mastery (which concerns the strategies one employs in the service of regulating one's mental states) (Semerari et al., 2003). In terms of understanding one's own mind, individuals with schizophrenia demonstrate deficits in self-attributing their own actions (Franck et al., 2001), which relate to symptoms such as passivity phenomenon. They also demonstrate inappropriate conviction in their beliefs (Martin et al., 2010), which relates to the delusions and the lack of insight that are characteristic of psychotic disorders. Additionally, individuals with schizophrenia exhibit impairments in understanding others' minds, being less able to accurately infer the affective states of other people when compared with healthy controls (Lee et al., 2011). This is likely to contribute to the poor social functioning that is commonly associated with this illness (Lysaker et al., 2007), which in turn is likely to be compounded by the impoverished social schemata that are associated with deficits in the mastery domain of metacognition seen in schizophrenia (Lysaker et al., 2010).

Mastery was also found to be significantly impaired individuals with personality disorders (Carcione et al., 2011), while metacognition in general has also been shown to be adversely affected in other psychiatric disorders such as PTSD (Bennett & Wells, 2010), and obsessive-compulsive disorder (OCD) and panic disorder (Cucchi et al., 2012), where beliefs about beliefs are especially important. For instance, in OCD, metacognitive deficits, such as low confidence in memory, may be linked to checking behaviors, either as a cause (Exner et al., 2009) or a consequence (Radomsky et al., 2014). In addition, beliefs about the controllability of thoughts are central to the condition (Purdon & Clark, 1999).

3.5 | Emotional regulation

Significant impairment of emotional regulation is a cardinal feature of borderline personality disorder and one which is associated with considerable suffering (Bohus et al., 2021). In addition to its deleterious impact on sustaining

relationships (Miano et al., 2017), increased impairment in this domain is associated with an increased likelihood of suicide attempts in this population (Mirkovic et al., 2021). A cross-sectional study demonstrated that it is also a feature of this condition that shows little variation over time, with people aged 40–59 years still being similarly impaired to those aged 18–25 years (Martino et al., 2020). The impairment of emotional regulation in borderline personality disorder has been shown to be compounded by comorbid bipolar affective disorder (Bayes et al., 2016). Emotional regulation is also found to be impaired in individuals with a solitary diagnosis of bipolar affective disorder, albeit to a lesser degree than in those with a single diagnosis of borderline personality disorder (Bayes et al., 2016).

Other psychiatric disorders in which emotional regulation has also been shown to be adversely affected include autism spectrum disorder (ASD) (Joshi et al., 2018), substance misuse disorder (Parolin et al., 2017), and attention deficit hyperactivity disorder (El Archi et al., 2023). Different disorders may affect different mechanisms of emotional regulation. For example, it has been shown that in social anxiety there is an overreliance in expressive suppression of emotions, while in major depressive disorder cognitive reappraisal is underutilized (Dryman & Heimberg, 2018).

3.6 | Autobiographical memory

A central component of self-awareness is autobiographical memory. Deficits in autobiographical memory are seen in a range of neuropsychiatric conditions and are a characteristic feature of most forms of dementia. In Alzheimer's disease (AD), deficits in episodic and semantic memory contribute to impaired autobiographical self-awareness (Mograbi et al., 2021). Deficits in autonoetic consciousness, the subjective conscious experience that accompanies episodic memory retrieval and involves “mental time travel” back to the event, autobiographical knowledge and a sense of continuity of the self occurs at the mild–moderate stages of AD (Huntley et al., 2021). People with AD may also be impaired in updating and integrating autobiographical information into their narrative sense of self, leading to a lack of self-awareness of current deficits (Lenzoni et al., 2020).

Other common neuropsychiatric conditions also demonstrate alterations in autobiographical memory. Autobiographical dysfunction may be both a cause and consequence of depression (Talarowska et al., 2016), with autobiographical memory recall identified as more generalized and less specific in recurrent depressive disorders (Hamlat et al., 2014; Hermans et al., 2008). People with depression are less able to access specific autobiographical memories in response to positive and neutral stimuli than controls (Young et al., 2012) or recall and reconstruct details of events, particularly those that were positive and may not easily integrate with a negative sense of self (Lemogne et al., 2006). Similarly, depressed bipolar patients report memories with fewer perceptual and time integration details than those in euthymia or mania (Da Silva et al., 2021). Over-general recall of autobiographical memory also occurs in patients with schizophrenia, with less retrieval of specific memories, which is associated with people with schizophrenia experiencing abnormalities in their concept of self (McLeod et al., 2006; Zhang et al., 2019).

4 | ALTERNATIVE FRAMEWORKS AND FUTURE RESEARCH DIRECTIONS

The evidence reviewed here indicates both the pervasiveness and diversity of self-awareness alterations in clinical conditions. The pattern of changes, within and across clinical groups, may help in developing cognitive frameworks to understand self-awareness. Here a PC approach may be particularly helpful to make sense of the data.

An initial perspective is to consider that, in some conditions, difficulties may arise in how higher-order processes constrain incoming bottom-up information. For instance, in depression and anxiety, changes in beliefs may affect allocation of attention and the interpretation of interoceptive signals. This may happen even across self-awareness processes. For instance, it is possible that emotional regulation beliefs impact directly on interoceptive processing (Critchley & Garfinkel, 2017).

This model, with alterations in higher-order constructs leading to changes in lower-level processes, may be helpful in understanding conditions such as OCD, in which beliefs and thoughts about thoughts are central (Myers & Wells, 2005). It is possible, however, that the priors themselves are either reinforced or formed on the basis of distorted information. This would lead to two different perspectives: (1) concomitant changes in low- and higher-order processes, with an altered interpretation of signals reinforcing beliefs and beliefs constraining signals, in a cycle; (2) a primarily bottom-up origin for difficulties, with beliefs being formed from disrupted incoming information. Still in the example of

OCD, it is possible that limbic hyperactivation and disrupted emotional processing lead to the altered thinking patterns of the condition (Simon et al., 2014; Thorsen et al., 2018).

In conditions such as schizophrenia, problems seem to appear across self-awareness processes, for instance with increased interoceptive sensitivity, poor bodily integration and decreased agency, but also metacognitive distortions. In this case, it is harder to suggest a straightforward chain of causality. These findings may suggest either a concomitant origin or a spurious correlation; in the latter, other changes, central to the condition, would lead to the profile of self-awareness alterations, which can either be considered an epiphenomenon or a reinforcing factor in the subjective experience of patients.

The expansion of a transdiagnostic approach to study self-awareness may deliver results with relevant clinical and theoretical implications. For instance, exploring how different self-awareness processes are affected across disorders may reveal shared diagnostic characteristics, going beyond a categorical approach. This is consistent with recent efforts to describe psychopathology in terms of dimensional features (Cuthbert, 2014), and may indicate relevant targets for intervention, both for pharmacological and psychosocial treatments. Although there is already some existing evidence, such as the studies reviewed in the previous section, further data collection may assist in determining mechanisms behind self-awareness in health and illness.

Conversely, exploring different self-awareness processes within single clinical conditions may not only provide important resources for interventions, but also shed light into the cognitive mechanisms of self-awareness. For example, investigating how bottom-up and top-down processes from different self-awareness subdomains interact, may indicate if there is a hierarchical relationship between these subdomains, with some capacities being “nested” in others (Feinberg, 2011). Exploring a variety of self-awareness processes within the same condition may also help clarify the interaction between self-awareness levels and the mechanisms leading to the emergence of subjective experience.

For this to happen, methodological advances in how to quantify self-awareness are needed. Different approaches have been used depending on the specific self-awareness process, including questionnaires (e.g., MAIA; Mehling et al., 2012), behavioral tests (e.g., agency, Imaizumi & Tanno, 2019) and physiological measures (e.g., measurement of event-related potentials for error monitoring, Lenzoni et al., 2021; resting-state magnetic resonance imaging; Ham et al., 2013), but comparatively fewer studies have integrated methods across these measurement levels. For example, the development of “self-awareness test batteries,” including questionnaires and behavioral measures, which can be coupled with physiological data, may elucidate the relationship between self-awareness levels and processes. Privileging ecological validity is another important focus for the development of new measurements, for instance with the use of virtual reality, which can provide immersive environments maintaining experimental control. Additionally, the development of new methods that can tap into the level of self-awareness of people with disorders of consciousness (e.g., persistent vegetative state, coma) is particularly relevant from an ethical point of view, for instance impacting on end-of-life decision-making.

Elucidating the neurochemical basis of self-awareness may have direct implications, not only for our understanding of this phenomenon, but also for diagnosis and treatment of neuropsychiatric conditions. An approach would be to identify which neurotransmitters are implicated in alterations of self-awareness. For example, the role of dopamine in agency has been suggested, with higher accessibility of dopamine potentially increasing sense of agency (Render & Jansen, 2019). Such findings would indicate targets for intervention, new potential therapies and highlight clinical groups who are more vulnerable to these changes. Another avenue of research would be to explore substances with traditional use that more recently have been incorporated into mainstream treatments. For example, psilocybin has been known to lead to changes in the sense of self, with users typically reporting “ego-dissolution” (a reduction of self-reference and a disintegration in the boundaries between the self and the world), in both negative and positive experiences with psilocybin (Bienemann et al., 2020; 2023). Considering the boom in research on psychedelics, investigating self-awareness may help bringing some of these compounds into clinical practice, also providing resources for a cognitive neuroscience of self-awareness.

Finally, exploring self-awareness across species may be crucial to understand the origins of this phenomenon and its evolutionary pathways. A comparative approach may indicate the extent to which levels of self-awareness are present in other species, as well as the particularities of certain processes in specific clades (Lage et al., 2022). Expanding our knowledge about the level of self-awareness of other species may have an important impact in our relationship with them. This evolutionary perspective may also shed light into another frontier of human knowledge, namely the capacity of devices to become self-aware. Various attempts have been made to explore this issue (Chella et al., 2019; McCarthy, 2002; Scheutz, 2014), for example, investigating the extent to which inner speech may be linked to self-awareness in robots (Chella et al., 2020). This could help revealing if inner speech is sufficient for the development of self-awareness (or, rather, which type of self-awareness may emerge from such an

attempt), while an evolutionary approach may indicate whether this construct is necessary or not. By exploring self-awareness in devices and increasing our understanding of the biological conditions in which self-awareness emerges, we will be more equipped to develop models and simulations of the phenomenon, also increasing our knowledge about the relevance of the material support grounding self-awareness, as well as its potential to be instantiated elsewhere.

5 | CONCLUSIONS

The search for self-awareness has been a journey for our species, with answers to this question being traditionally provided by religion and philosophy. A cognitive neuroscience of self-awareness has only been emerging in the past few decades, but growing evidence has been shedding light into the phenomenon, not least in the case of neuropsychiatric conditions, as reviewed here. It is expected that the expansion of this field will have implications for theoretical, clinical, and ethical perspectives.

One important limitation of the current review refers to the lack of evidence from across conditions, with few studies directly comparing different disorders. Although such studies are indeed scarce, we hope that the current work will motivate researchers to pursue this further, not least because this transdiagnostic approach may also be relevant clinically. Additionally, some may argue against the attempt to bring processes as diverse as interoception and metacognition under the umbrella of self-awareness. Indeed, part of the scientific endeavor is grounded in breaking up reality into smaller components, in order to better understand its mechanisms, then reassembling its pieces. The approach indicated here does not replace the need for the examination of individual self-awareness processes, but suggests that exploring these together may reveal shared mechanisms and interactions that could be harder to identify in isolation.

Our cognitive faculties have evolved with the sole purpose of keeping us alive, helping us to navigate the environment and identify internal demands and their level of satisfaction. Within this perspective, cognition is self-serving by definition. By exploring the mechanisms of self-awareness, a finer understanding of brain function and cognitive processing may be achieved, helping us to answer the perennial question of who we are, both from a mechanistic and a subjective perspective.

AUTHOR CONTRIBUTIONS

Daniel Mograbi: Conceptualization (equal); methodology (equal); writing – original draft (lead); writing – review and editing (lead). **Simon Hall:** Conceptualization (supporting); writing – original draft (equal). **Beatriz Arantes:** Writing – original draft (supporting); writing – review and editing (equal). **Jonathan Huntley:** Conceptualization (equal); methodology (equal); writing – original draft (lead); writing – review and editing (lead).

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

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